

# GenMol / UDLM engineering study

Complete V5 temperature screen and V6 Gibbs comparison | source snapshot 2026-09-07T10:09:32.675735+00:00

**No superiority established. No formal promotion or final candidate lock.** The highest observed repaired quality across the 18 engineering configurations is 57.03% (V6, s\_t050\_gibbs), versus 85.80% for the frozen local MDLM baseline. This descriptive maximum is selected from a small screen; it is not a confirmatory estimate.

| Evidence                   | Configurations / runs / requested samples                             | Seed and compute budget                                                               |
|----------------------------|-----------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| V5 temperature screen      | 12 configurations; 24 runs; 1,536 samples, all independently rescored | R/S/E x temperatures 0.50, 0.70, 0.85, 1.00; seeds 1200/1201; 128 predictor NFEs      |
| V6 Gibbs comparison        | 6 configurations; 12 runs; 768 samples, all independently rescored    | R/S/E x predictor/Gibbs; seeds 1300/1301; temperature 0.50; 128 total NFEs            |
| Frozen local MDLM baseline | 1 configuration; 3 runs; 3,000 samples                                | Seeds 0/1/2; released confidence sampling; inference and training compute not matched |

**128 samples per engineering configuration versus 3,000 baseline samples.** Every engineering result averages two 64-sample seeds. All 2,304 V5+V6 requests remain disclosed; they are not pooled to estimate a single model configuration. V6 temperature 0.50 was chosen using partial V5 results, so it is an informed engineering choice, not a separately selected global optimum.

**Training and checkpoints.** Each R/S/E model starts from the same 50,000-update MDLM EMA and adds 1,000 updates at global batch 16 (16,000 example exposures; one GPU; microbatch 2; accumulation 8; seed 17). All three retain their original checkpoint bytes. The local MDLM training used batch 2,046 for 50,000 updates (102.3 million requested exposures); the paper used batch 2,048. A matched extra-update MDLM control is still absent. Scheduler and FiLM choices were selected on E loss, then shared across arms.

**Arm definitions.** R uses released uniform UDLM with its known loss/forward schedule mismatch; S repairs that schedule using uniform categorical diffusion; E uses the same repaired process and empirical SAFE prior with a 0.0002 uniform mixture. None is a published reproduction of a fully optimized UDLM molecular model.

**Metrics and uncertainty.** Validity = valid/requested; uniqueness = unique/valid; quality = distinct valid molecules with QED >= 0.6 and SA <= 4, divided by requested samples. QED summarizes drug-likeness; SA is the synthetic-accessibility score (lower is easier). Diversity is the released PyTDC fingerprint diversity on unique valid molecules. Means and sample SD average seeds equally; diversity is not pooled. Two-seed SD is fragile, not a confidence interval or significance test; configuration selection and molecular dependence add uncertainty.

**Decoding and next work.** Repaired metrics use SAFE fix=True and select the largest component by SMILES string length. Strict metrics use fix=False plus RDKit validation, without repair or component selection. Final UDLM seeds 0/1/2 remain reserved (the historical baseline already used those seeds). CE denoising with checkpoint-bound CE-to-LOO conversion is implemented and CPU-tested, but remains untrained and has no molecular results in this report. The archived V4 failure remains failed and contributes no ranked configuration.

# Every configuration: mean +/- sample SD across two seeds

V/U/Q are percentages; D is diversity. P = predictor backbone evaluation; G = fresh Gibbs evaluation. All engineering rows use 128 samples/configuration and top-p=1.00. Baseline rows are context, not matched tests.

| Study / arm | T    | Kernel   | Repair V%       | Repair U%       | Repair Q%       | Repair D        | Strict V%       | Strict U%       | Strict Q%      | Strict D        |
|-------------|------|----------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|----------------|-----------------|
| V5 / R      | 0.50 | 128P     | 99.22 +/- 1.10  | 100.00 +/- 0.00 | 48.44 +/- 4.42  | 0.881 +/- 0.001 | 63.28 +/- 5.52  | 100.00 +/- 0.00 | 29.69 +/- 4.42 | 0.869 +/- 0.001 |
| V5 / R      | 0.70 | 128P     | 97.66 +/- 3.31  | 100.00 +/- 0.00 | 44.53 +/- 1.10  | 0.894 +/- 0.002 | 46.09 +/- 1.10  | 100.00 +/- 0.00 | 19.53 +/- 5.52 | 0.875 +/- 0.002 |
| V5 / R      | 0.85 | 128P     | 96.88 +/- 2.21  | 99.21 +/- 1.12  | 35.94 +/- 6.63  | 0.911 +/- 0.002 | 39.84 +/- 3.31  | 100.00 +/- 0.00 | 21.09 +/- 5.52 | 0.877 +/- 0.010 |
| V5 / R      | 1.00 | 128P     | 93.75 +/- 0.00  | 99.17 +/- 1.18  | 30.47 +/- 1.10  | 0.911 +/- 0.005 | 37.50 +/- 6.63  | 100.00 +/- 0.00 | 14.06 +/- 2.21 | 0.881 +/- 0.003 |
| V5 / S      | 0.50 | 128P     | 97.66 +/- 1.10  | 99.19 +/- 1.14  | 41.41 +/- 14.36 | 0.897 +/- 0.004 | 60.16 +/- 1.10  | 100.00 +/- 0.00 | 23.44 +/- 8.84 | 0.878 +/- 0.004 |
| V5 / S      | 0.70 | 128P     | 97.66 +/- 1.10  | 100.00 +/- 0.00 | 42.19 +/- 19.89 | 0.898 +/- 0.006 | 42.97 +/- 1.10  | 100.00 +/- 0.00 | 21.09 +/- 1.10 | 0.864 +/- 0.004 |
| V5 / S      | 0.85 | 128P     | 96.09 +/- 1.10  | 100.00 +/- 0.00 | 50.78 +/- 5.52  | 0.904 +/- 0.004 | 39.84 +/- 5.52  | 100.00 +/- 0.00 | 24.22 +/- 3.31 | 0.871 +/- 0.004 |
| V5 / S      | 1.00 | 128P     | 92.97 +/- 7.73  | 98.41 +/- 2.24  | 32.03 +/- 1.10  | 0.924 +/- 0.005 | 25.00 +/- 0.00  | 100.00 +/- 0.00 | 11.72 +/- 1.10 | 0.886 +/- 0.002 |
| V5 / E      | 0.50 | 128P     | 97.66 +/- 1.10  | 100.00 +/- 0.00 | 45.31 +/- 6.63  | 0.892 +/- 0.006 | 47.66 +/- 1.10  | 100.00 +/- 0.00 | 22.66 +/- 5.52 | 0.878 +/- 0.002 |
| V5 / E      | 0.70 | 128P     | 98.44 +/- 0.00  | 99.21 +/- 1.12  | 45.31 +/- 11.05 | 0.899 +/- 0.003 | 40.62 +/- 11.05 | 100.00 +/- 0.00 | 21.09 +/- 9.94 | 0.872 +/- 0.015 |
| V5 / E      | 0.85 | 128P     | 94.53 +/- 1.10  | 96.68 +/- 2.38  | 39.84 +/- 3.31  | 0.904 +/- 0.005 | 32.03 +/- 3.31  | 100.00 +/- 0.00 | 13.28 +/- 3.31 | 0.883 +/- 0.010 |
| V5 / E      | 1.00 | 128P     | 93.75 +/- 4.42  | 99.14 +/- 1.22  | 35.94 +/- 6.63  | 0.920 +/- 0.012 | 17.97 +/- 3.31  | 100.00 +/- 0.00 | 9.38 +/- 4.42  | 0.884 +/- 0.007 |
| V6 / R      | 0.50 | 128P     | 100.00 +/- 0.00 | 98.44 +/- 2.21  | 53.91 +/- 5.52  | 0.885 +/- 0.007 | 65.62 +/- 8.84  | 100.00 +/- 0.00 | 36.72 +/- 3.31 | 0.874 +/- 0.001 |
| V6 / R      | 0.50 | 64P+64G  | 97.66 +/- 1.10  | 100.00 +/- 0.00 | 47.66 +/- 3.31  | 0.893 +/- 0.012 | 61.72 +/- 7.73  | 100.00 +/- 0.00 | 32.03 +/- 5.52 | 0.877 +/- 0.008 |
| V6 / S      | 0.50 | 128P     | 98.44 +/- 2.21  | 99.19 +/- 1.14  | 54.69 +/- 4.42  | 0.882 +/- 0.006 | 61.72 +/- 1.10  | 100.00 +/- 0.00 | 33.59 +/- 1.10 | 0.866 +/- 0.007 |
| V6 / S      | 0.50 | 64P+64G  | 98.44 +/- 2.21  | 97.61 +/- 1.18  | 57.03 +/- 1.10  | 0.892 +/- 0.005 | 55.47 +/- 7.73  | 100.00 +/- 0.00 | 36.72 +/- 1.10 | 0.878 +/- 0.004 |
| V6 / E      | 0.50 | 128P     | 96.88 +/- 0.00  | 99.19 +/- 1.14  | 46.88 +/- 8.84  | 0.896 +/- 0.010 | 53.91 +/- 5.52  | 100.00 +/- 0.00 | 31.25 +/- 2.21 | 0.887 +/- 0.006 |
| V6 / E      | 0.50 | 64P+64G  | 98.44 +/- 2.21  | 98.41 +/- 0.04  | 51.56 +/- 6.63  | 0.889 +/- 0.003 | 55.47 +/- 9.94  | 100.00 +/- 0.00 | 29.69 +/- 0.00 | 0.874 +/- 0.003 |
| Local MDLM  | 0.50 | released | 100.00 +/- 0.00 | 99.87 +/- 0.06  | 85.80 +/- 1.23  | 0.823 +/- 0.001 | 98.80 +/- 0.20  | 99.87 +/- 0.06  | 84.77 +/- 1.22 | 0.822 +/- 0.002 |

The local MDLM row averages three 1,000-sample seeds. Published GenMol V1 repaired reference: validity 100.0%, uniqueness 99.7%, quality 84.6%, diversity 0.818 (paper run means). Hardware, training, data provenance and sample counts differ; higher diversity alone does not establish better molecules.

# Within-arm V6 Gibbs deltas at fixed total NFE

Treatment minus predictor control, pairing seed 1300 with 1300 and 1301 with 1301 at the same checkpoint. Control: 128 predictor evaluations. Treatment: 64 predictor transitions + 64 fresh random-scan Gibbs updates. The paired RNG streams diverge because Gibbs changes draw counts. These are descriptive paired-seed deltas, not per-molecule paired effects or a confidence interval. Temperature is 0.50 in every pair.

| Arm | Decoding | Delta validity (pp) | Delta uniqueness (pp) | Delta quality (pp) | Delta diversity  |
|-----|----------|---------------------|-----------------------|--------------------|------------------|
| R   | repaired | -2.34 +/- 1.10      | +1.56 +/- 2.21        | -6.25 +/- 2.21     | +0.008 +/- 0.019 |
| R   | strict   | -3.91 +/- 16.57     | +0.00 +/- 0.00        | -4.69 +/- 2.21     | +0.003 +/- 0.009 |
| S   | repaired | +0.00 +/- 4.42      | -1.59 +/- 2.32        | +2.34 +/- 3.31     | +0.010 +/- 0.001 |
| S   | strict   | -6.25 +/- 6.63      | +0.00 +/- 0.00        | +3.12 +/- 0.00     | +0.012 +/- 0.011 |
| E   | repaired | +1.56 +/- 2.21      | -0.78 +/- 1.18        | +4.69 +/- 2.21     | -0.007 +/- 0.007 |
| E   | strict   | +1.56 +/- 15.47     | +0.00 +/- 0.00        | -1.56 +/- 2.21     | -0.013 +/- 0.003 |

The means show mixed effects: E improves repaired quality but loses strict quality; S improves both quality means while strict validity falls; R declines in both quality and validity branches. There is no universal Gibbs gain. V5 and V6 use different seeds, so their differences are not paired comparisons.

A random-scan Gibbs step resamples one uniformly chosen editable coordinate per sequence and may leave its value unchanged. Exact stationarity requires true compatible LOO conditionals and untempered sampling; these learned predictions at temperature 0.50 form an approximate corrector. This comparison also trades away half the predictor transitions, so it evaluates that allocation, not adding free extra compute.

| Checkpoint / budget                   | SHA-256 (original source bytes; not reserialized)                |
|---------------------------------------|------------------------------------------------------------------|
| MDLM 50k EMA warm-start source        | 8d00aa47b02f64bf39ff6b0b2e786f213587366fc2c3d29712a00f3f84108dd6 |
| R: +1,000 updates / +16,000 exposures | d0310d2e2402043ab9ab6a99268263581a35d60fb2262ddd988ea38cd6f6395c |
| S: +1,000 updates / +16,000 exposures | 100f467b94766f2c87cc734398c8590bd14a1c8e0722ce31dbca7b446d986e72 |
| E: +1,000 updates / +16,000 exposures | dce870e8d63453f73c428b9115f33556f67a21d45788d3112c2777ea82623005 |

**Artifact trail.** The input manifest hashes both report JSON/CSV/PDF files, all 36 run summaries and raw CSVs, both protocols and their configuration bytes, the three continuation training summaries, the local baseline aggregate and its verified PDF, and this generator. Checkpoint identities are inherited from the independently validated generation/training records; this overview does not reload multi-gigabyte checkpoints.

**Unmodified appendices follow:** complete independently rescored V5 report; complete independently rescored V6 report; frozen local MDLM baseline report. They retain original page labels and historical context. The companion CSV/JSON contain all configuration means, SDs, paired deltas, hashes and original identifiers. No significance test, superiority declaration, formal promotion, or reserved-seed UDLM generation occurred here.